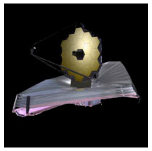

7793 - A Deep Dive Spectroscopic Study of an Auroral Emitting World

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NIRSPEC				
	2	1935-1546	NIRSpec Fixed Slit Spectroscopy	(1) J193518.59-154620.3
	5	W1935-1546	NIRSpec Fixed Slit Spectroscopy	(1) J193518.59-154620.3
MIRI MRS				
	6	W1935_MRS	MIRI Medium Resolution Spectroscopy	(1) J193518.59-154620.3

ABSTRACT

In this JWST proposal, we will obtain NIRSpec and MRS spectra of the cold brown dwarf WISEP J193518.59-154620.3 in order to characterize a spectroscopic example of an auroral emitting extrasolar world. The methane emission seen in a Cycle 1 program on this source is also found in the G395H spectra of planetary objects (e.g. Jupiter, Uranus, Pluto) and expected in extrasolar planets therefore the detailed characterization we propose crosses substellar, exoplanet, and planetary sub-fields. At a Spitzer [3.6] band mag of ~18.1, W1935 is impossibly faint for ground based observatories therefore JWST is the only facility that can complete our science agenda. The power of this proposal will be in answering the following questions: (1) Is there a forest of CH4 and H3+ emission in W1935 like that seen in solar system objects (e.g. Jupiter, Uranus, Neptune) observed with the same NIRSpec G395H set-up? (2) Do the W1935 emission characteristics extend into the mid infrared where a retrieved model predicts we will see more emission from CH4 and NH3+? (3) Will modeling the spectrum of W1935 lead to conclusive evidence about whether the 300K temperature inversion in local thermodynamic equilibirum (LTE) is the correct interpretation of the cause of the emission or if a non-LTE (NLTE) process linked somehow to a magnetic interaction with the atmosphere might be at play? The NIRSpec and MRS spectra of W1935 would be the first ever spectroscopic study of an auroral emitting world beyond our own and would showcase the power of JWST at atmospheric characterization crossing from extrasolar worlds into the solar system.

OBSERVING DESCRIPTION

This proposal is to obtain NIRSpec G395H spectra along with MIRI MRS spectra for the cold brown dwarf WISEP J193518.59-154620.3 (W1935). W1935 has a cycle 1 G395H spectra which demonstrated methane emission and this proposal will re-visit to acquire further signal on the source as well as extended wavelength coverage to fully characterize its spectral energy distribution. W1935 has a precise parallax and proper motion for accuracy in positioning the source on the slit. We will acquire the target using the default WATA mode with the CLEAR filter for NIRSpec and the F1000W filter for MIRI spectroscopy. We will dither along the slit using a 3-point pattern for NIRSpec G395H, a 5-point pattern for prism and a 4-point pattern for MIRI MRS. We use the NRSRAPID readout for NIRSPec and SLOWR1 readout for MIRI.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J193518.59-154620.3	RA: 19 35 18.7380 (293.8280750d) Dec: -15 46 20.33 (-15.77231d) Equinox: J2000	Proper Motion RA: 290.2 mas/yr Proper Motion Dec: 43.1 mas/yr Parallax: .0693" Epoch of Position: 2022.71679665	

Comments: This is a copy of Target 1
Category=Star
Description=[Brown dwarfs]
Extended=NO

Observation

Proposal 7793, Observation 2: 1935-1546

Tue Jul 01 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Comments: This is a duplicate of failed observation 01

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J193518.59-154620.3	RA: 19 35 18.7380 (293.8280750d) Dec: -15 46 20.33 (-15.77231d) Equinox: J2000	Proper Motion RA: 290.2 mas/yr Proper Motion Dec: 43.1 mas/yr Parallax: .0693" Epoch of Position: 2022.71679665	

Comments: This is a copy of Target 1
Category=Star
Description=[Brown dwarfs]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	1 J193518.59-154620.3	WATA	SUB2048	CLEAR	NRSRAPID	3	1	1	3.628	226182

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395H/F290LP	S200A1	NRSRAPID	300	13	1	NONE	3	39	18290.161	226182

Special Requirements	
	Group Observations 2, 5, 6, Non-interruptible

Observation

Proposal 7793, Observation 5: W1935-1546

Tue Jul 01 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J193518.59-154620.3	RA: 19 35 18.7380 (293.8280750d) Dec: -15 46 20.33 (-15.77231d) Equinox: J2000	Proper Motion RA: 290.2 mas/yr Proper Motion Dec: 43.1 mas/yr Parallax: .0693" Epoch of Position: 2022.71679665	

Comments: This is a copy of Target 1
Category=Star
Description=[Brown dwarfs]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	1 J193518.59-154620.3	WATA	SUB2048	CLEAR	NRSRAPID	3	1	1	3.628	226182

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	5	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	PRISM/CLEAR	S200A1	NRSRAPID	75	5	1	NONE	5	25	2960.712	226182

Special Requirements	
	Group Observations 2, 5, 6, Non-interruptible

Observation

Proposal 7793, Observation 6: W1935_MRS

Tue Jul 01 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 6:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J193518.59-154620.3	RA: 19 35 18.7380 (293.8280750d) Dec: -15 46 20.33 (-15.77231d) Equinox: J2000	Proper Motion RA: 290.2 mas/yr Proper Motion Dec: 43.1 mas/yr Parallax: .0693" Epoch of Position: 2022.71679665	

Comments: This is a copy of Target 1
Category=Star
Description=[Brown dwarfs]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	1 J193518.59-154620.3	F1500W	FASTGRPAVG	4	1	1	44.401	226182

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	NO	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		SLOWR1	27	12	1	Dither 1	4	48	32012.493	226182
1	SHORT(A)	MRSSHORT		SLOWR1	27	12	1	Dither 1	4	48	32012.493	226182
2	MEDIUM(B)	MRSLONG		SLOWR1	27	12	1	Dither 1	4	48	32012.493	226182
2	MEDIUM(B)	MRSSHORT		SLOWR1	27	12	1	Dither 1	4	48	32012.493	226182
3	LONG(C)	MRSLONG		SLOWR1	27	12	1	Dither 1	4	48	32012.493	226182
3	LONG(C)	MRSSHORT		SLOWR1	27	12	1	Dither 1	4	48	32012.493	226182

Special Requirements	
	Group Visits within 53.0 Days Visits Same PA Group Observations 2, 5, 6, Non-interruptible